

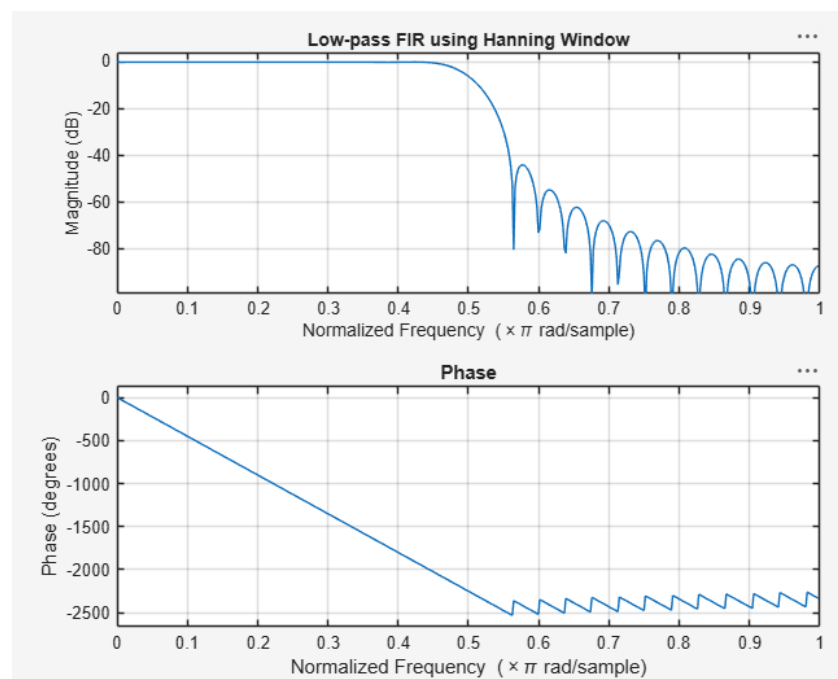
Expt 4 : Design of Low Pass Filter using Windowing Techniques

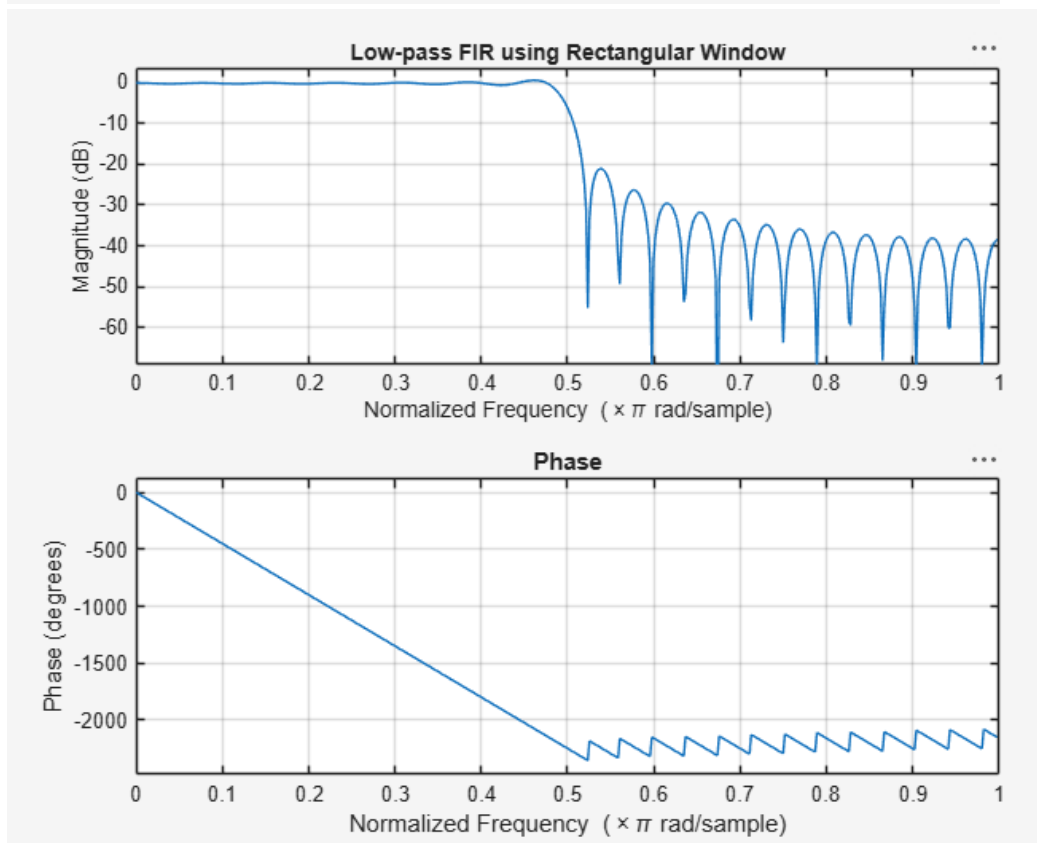
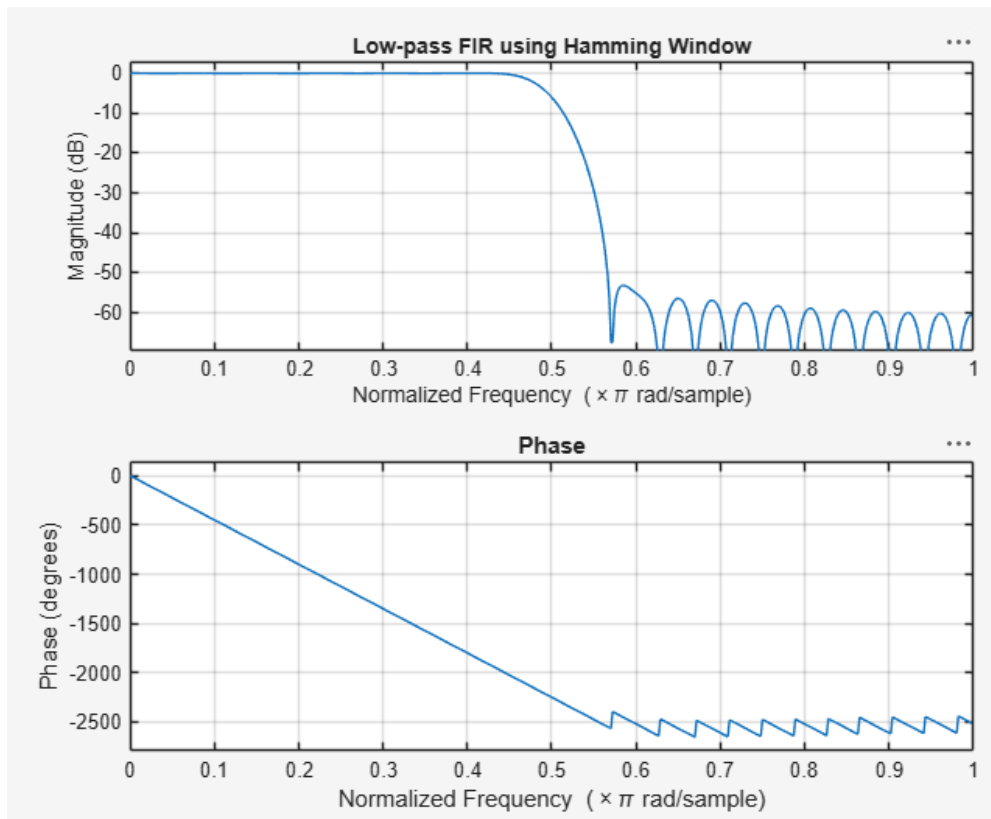
4..1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique

Program :

```
/MATLAB Drive/Dsp_4a.m
2 clear;
3 close all;
4 %% --- Input Parameters ---
5 fp = 1000; % Passband cutoff (Hz)
6 fs = 4000; % Sampling frequency (Hz)
7 N = 50; % Filter order (even number preferred)
8
9 wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
10 %% --- 1. FIR Low Pass using Rectangular Window ---
11 b_rect = fir1(N, wc, 'low', rectwin(N+1));
12 disp('--- Rectangular Window Coefficients ---');
13 disp(b_rect');
14 %% --- 2. FIR Low Pass using Hamming Window ---
15 b_hamm = fir1(N, wc, 'low', hamming(N+1));
16 disp('--- Hamming Window Coefficients ---');
17 disp(b_hamm');
18 %% --- 3. FIR Low Pass using Hanning Window ---
19 b_hann = fir1(N, wc, 'low', hanning(N+1));
20 disp('--- Hanning Window Coefficients ---');
21 disp(b_hann');
22 %% --- Plot Frequency Responses ---
23 figure;
24 freqz(b_rect,1);
25 title('Low-pass FIR using Rectangular Window');
26 figure;
27 freqz(b_hamm,1);
28 title('Low-pass FIR using Hamming Window');
29 figure;
30 freqz(b_hann,1);
31 title('Low-pass FIR using Hanning Window');
```

Output :





4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

Program:

```
/MATLAB Drive/Dsp_4b.m

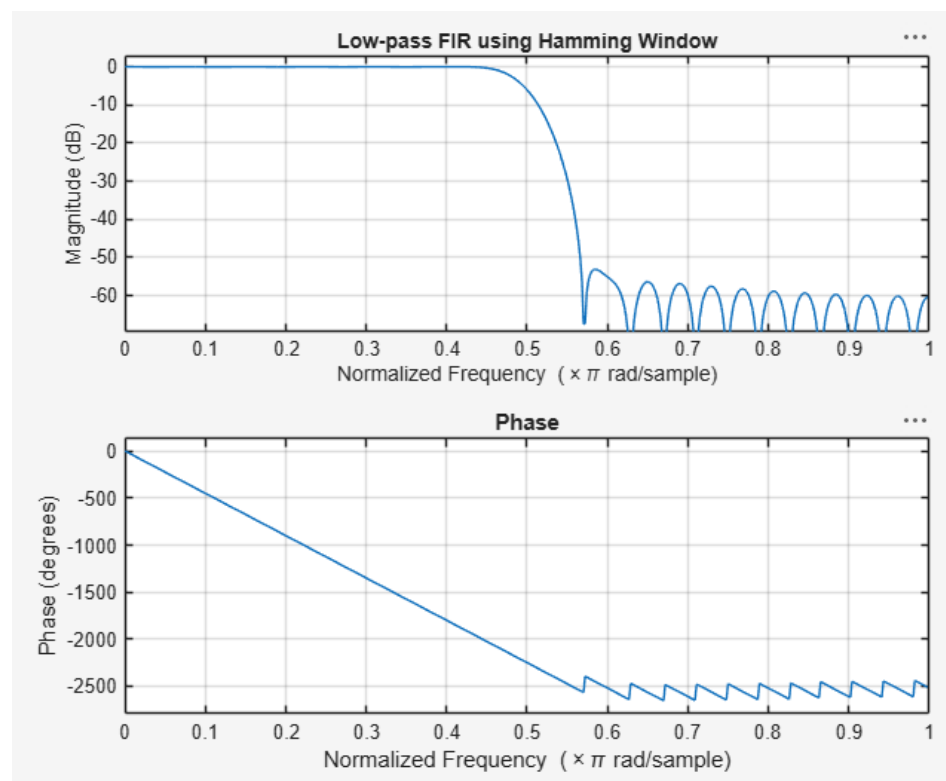
1  clc;
2  clear;
3  close all;

4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

9  %% --- FIR Low Pass using Hamming Window ---
10
11  b_hamm = fir1(N, wc, 'low', hamming(N+1));
12  disp('--- Hamming Window Coefficients ---');
13  disp(b_hamm');

14  %% --- Plot Frequency Responses ---
15  figure;
16  freqz(b_hamm,1);
17  title('Low-pass FIR using Hamming Window');
```

Output:



4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique

Program:

/MATLAB Drive/Dsp_4c.m

```
1  clc;
2  clear;
3  close all;
4  %% --- Input Parameters ---
5  fp = 1000; % Passband cutoff (Hz)
6  fs = 4000; % Sampling frequency (Hz)
7  N = 50; % Filter order (even number preferred)
8  wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9
10 %% --- FIR Low Pass using Hanning Window ---
11 b_hann = fir1(N, wc, 'low', hanning(N+1));
12 disp('--- Hanning Window Coefficients ---');
13 disp(b_hann');
14 %% --- Plot Frequency Responses ---
15 figure;
16 freqz(b_hann,1);
17 title('Low-pass FIR using Hanning Window');
```

Output:

